Lesson notes · for you

A picture is a grid of numbers

University, vision 1

Time
45 min

Materials
A phone with Darsi; Paper for one hand-worked step; The slides from the kit

Learning goals

- One number per cell
- Three grids for colour

Activity: In the lab

1. Upright edges
2. Blur

Discussion

- What is the filter mixing?

Check yourself

- A filter does not know the name of what?

Try it this week
Sketch a 3×3 upright-edge filter and one output pixel by hand.

Worksheet

A picture is a grid of numbers

University, vision 1

Name _____ Class _____ Date _____

1 **A grey picture on a computer is…**

◯ A grid of numbers ◯ A list of object names ◯ One bit for the whole photo

2 **A 3×3 filter already knows it is looking at a face.**

◯ True ◯ False

3 **Match the filter.**

Upright edges	• • • •	Keep the strongest in a small block
Blur	• • • •	A pixel averaged with its neighbours
Pooling	• • • •	Walls and door sides

4 **Order one filter step.**

☐ Write that sum as the new pixel

☐ Multiply and add the nine cells

☐ Lay the 3×3 on the picture

✎ **What did the filter not know?**